

# Constraint-Violation Diagnostics for Hard-Negative Filtering in NQ RAG

UNIST CSE402 Default Project

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## 1 Research paper summary (max 2 pages)

- **Title:** QUEST: A Retrieval Dataset of Entity-Seeking Queries with Implicit Set Operations
- **Authors:** Chaitanya Malaviya, Peter Shaw, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova
- **Venue:** Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL), Volume 1: Long Papers
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**Background.** Many information-seeking queries cannot be answered by matching a question to a single topically related document. Users often express selective information needs that combine multiple constraints, such as entity type, location, time period, genre, or exclusion conditions. For example, a query such as “science-fiction films shot in England” requires finding entities that satisfy both a genre constraint and a filming-location constraint, while a query such as “shorebirds that are not sandpipers” additionally requires handling negation. Malaviya et al. argue that such queries implicitly require set operations such as intersection, union, and difference, but these phenomena are underrepresented in standard retrieval benchmarks. Existing question answering and retrieval datasets such as Natural Questions (Kwiatkowski et al., 2019) or MS MARCO (Nguyen et al., 2016) contain many realistic search queries, but they do not specifically evaluate whether a system can match multiple query constraints to textual evidence and return a complete set of satisfying entities. Structured knowledge-base question answering can handle some of these cases by grounding constraints to predefined predicates, but knowledge bases are expensive to maintain and often incomplete. The paper therefore asks whether text-based retrieval systems can directly match query constraints to evidence in documents without relying on structured KBs.

**Summary of contributions.** The main contribution of QUEST is a benchmark for evaluating retrieval systems on entity-seeking queries with implicit set operations. The dataset contains 3,357 natural language queries mapped to sets of Wikipedia entity documents. The

authors construct the dataset semi-automatically from Wikipedia categories: they first sample category names as atomic constraints, compose them using templates corresponding to set operations such as  $A \cap B$ ,  $A \cup B$ , and  $A \setminus B$ , and then ask crowdworkers to paraphrase and validate the resulting queries for fluency and naturalness. The dataset covers four domains: films, books, animals, and plants. This design makes the benchmark more controlled than arbitrary web search logs while still producing natural-language queries that resemble plausible search needs.

A second important contribution is the dataset’s evidence annotation. For a substantial subset of examples, crowdworkers do not merely mark an entity as relevant or irrelevant; they also highlight spans in the entity’s Wikipedia document that support the query constraints. This makes QUEST valuable not only for measuring retrieval accuracy, but also for studying whether systems can identify why a document satisfies a query. This is particularly relevant for interpretable retrieval and retrieval-augmented generation, where a system should ideally provide answers grounded in explicit evidence.

The paper also establishes baseline results using modern retrieval systems. The authors evaluate sparse BM25 retrieval, dense dual-encoder retrieval initialized from T5, and cross-attention classifiers that decide whether a candidate document is relevant to the query. The best baseline, a T5-Large dual encoder followed by a T5-Large classifier, achieves only 0.192 average F1 on the test set, showing that QUEST remains challenging for strong neural retrieval pipelines. The error analysis is especially informative: models struggle with intersection and difference operations, and many false positives satisfy only part of the query. For intersection queries, this suggests that systems may treat conjunction as if it were disjunction; for difference queries, the results suggest that systems are insufficiently sensitive to negated constraints. In this sense, the paper contributes not just a dataset, but also a diagnostic view of how retrieval systems fail on compositional natural-language constraints.

**Limitations and discussion.** Although QUEST is carefully designed, it has several limitations. First, the queries are derived from Wikipedia category names and semi-automatically composed using predefined templates. This gives the authors control over the set-operation structure, but it also means that QUEST is not an unbiased sample of real search-engine queries.

The authors validate fluency and naturalness through crowdsourcing, but the distribution of queries may still differ from naturally occurring user information needs.

Second, the benchmark is limited to four domains and to Wikipedia entity documents. These choices make annotation feasible and ensure relatively trusted source documents, but they restrict the diversity of language and evidence. The benchmark also assumes that relevant evidence can be found in an entity’s Wikipedia page. In practice, an entity may satisfy a query even if its Wikipedia page does not explicitly contain all supporting evidence. This creates a recall problem: systems may be penalized for retrieving entities that are truly relevant but missing from the category-derived answer set or lacking explicit textual evidence in the chosen document.

Third, QUEST evaluates retrieval of entity sets rather than end-to-end question answering or generation. This is appropriate for the paper’s goal, but it means that the benchmark does not directly evaluate whether a generative model can synthesize a final answer from retrieved evidence. For my purposes, this is both a limitation and a useful opportunity: the paper clearly identifies constraint-matching failures in retrieval, while my project can investigate a related problem in a RAG setting where retrieved passages are used to generate short answers.

Overall, I find the paper convincing because it isolates a real weakness of retrieval systems: models often retrieve documents that are topically related but fail to satisfy all constraints in the query. However, its conclusions should be interpreted as evidence about a controlled text-based retrieval setting, not as a complete solution to open-domain QA or RAG.

**Why this paper?** I chose this paper because my project is about improving retrieval-augmented question answering, and QUEST directly studies a failure mode that is relevant to RAG: topical relevance is not always enough. In the default project, the RAG system retrieves passages and then asks a language model to generate an answer from those passages. If the retrieved passages satisfy only part of the question, the generator may still produce a fluent but incorrect answer. QUEST provides a useful conceptual framework for thinking about this problem. Instead of treating each passage as simply relevant or irrelevant, the paper encourages a more structured view: which constraints in the question are supported, missing, or contradicted by the evidence?

This is closely related to the extension I plan to explore later in the project description. In particular, I am interested in using constraint-level diagnostics to filter hard-negative passages before final answer generation. QUEST is not itself a RAG paper, but it motivates why constraint-aware evidence selection may be useful for robust open-domain QA.

**Wider research context.** QUEST fits into a broader NLP question: can models understand the compositional structure of natural language well enough to re-

trieve and reason over evidence? Many NLP systems perform well when relevance can be approximated by semantic similarity, but language often expresses meaning through logical structure: conjunction, disjunction, negation, exclusion, and implicit constraints. QUEST shows that retrieval systems still struggle with this structure, especially when a document partially matches a query. This connects to fundamental issues in language representation. A good representation should not only encode that a query and a document are about similar topics; it should also preserve which specific constraints are satisfied.

The paper is also relevant to retrieval-augmented generation (Lewis et al., 2020). RAG systems rely on retrieved text as external knowledge, but the generator can only be as reliable as the evidence it receives. If a retriever returns passages that are topically similar but constraint-violating, the language model may generate unsupported or misleading answers. Therefore, the constraint-matching perspective in QUEST suggests a direction for more robust RAG: retrieved passages should be evaluated not only by scalar relevance, but also by whether they support the atomic constraints required by the question. Although QUEST focuses on entity-set retrieval, its analysis of partial constraint satisfaction and negation errors provides useful motivation for evidence filtering in open-domain question answering.

## 2 Project description (1-2 pages)

**Goal.** The main goal of this project is to investigate whether retrieval-augmented question answering can be made more robust by diagnosing which constraints in a question are satisfied, missing, or contradicted by each retrieved passage before answer generation. The scientific question I aim to answer is: *Can constraint-violation diagnostics improve RAG performance under hard-negative retrieval noise, compared to using BM25 relevance scores or prompt-based RAG alone?*

This goal is motivated by a limitation of standard RAG pipelines: retrieved passages may be topically relevant to the question but still fail to support the exact answer. For example, a passage about the first ascent of Mount Everest may be relevant to a question about the first ascent without supplemental oxygen, but it violates an important condition in the question. Such passages are hard negatives: they are semantically close enough to be retrieved, but misleading for answer generation. My chosen paper, QUEST (Malaviya et al., 2023), studies a related issue in entity retrieval: systems often fail when queries require multiple constraints or implicit set operations. I will adapt this constraint-matching perspective to the default project’s RAG setting by using constraint-level passage diagnostics as an evidence-filtering step before final generation.

The project will first complete the required default implementation of GPT-small and the baseline RAG system. My proposed extension will then focus on the

Llama-3.2-1B-Instruct RAG pipeline. As a stretch goal, if all retrieved passages miss the same core constraint, I will test a simple one-step targeted retrieval strategy using the missing constraint as an additional query signal.

**Task.** The main task is open-domain question answering on Natural Questions (NQ). Given a natural language question, the system retrieves passages from a Wikipedia passage corpus and generates a short answer. For example, given the input question “Who were the first people to climb Mount Everest without supplemental oxygen?”, the system should retrieve passages that support the specific condition “without supplemental oxygen” and output the correct answer, such as “Reinhold Messner and Peter Habeler.” The task is challenging because the retriever may return passages that are topically related but do not satisfy all constraints in the question.

In addition to the default QA task, my extension introduces an intermediate evidence-filtering task. Given a question and a retrieved passage, the system predicts whether each atomic question constraint is satisfied, missing, contradicted, or unrelated in the passage. This diagnostic output is then used to select the passages that are safest to include in the final RAG prompt.

**Data.** I will use the NQ open-domain QA dataset provided in the default project. The dataset contains 58,880 training examples and 6,515 development examples, with a Wikipedia corpus of 21,015,324 passages. Each example includes a question, answer strings, positive contexts, negative contexts, and hard negative contexts. The default RAG training uses question, answers, and positive contexts, while retrieval at evaluation time is handled by BM25 over the Wikipedia passage index.

For the proposed extension, I will also use the provided negative and hard negative contexts when available. These contexts are useful because they provide naturally occurring examples of passages that may be lexically or semantically close to the question but should not be used as final evidence. I do not plan to collect new data. Preprocessing will include normalizing answer strings for evaluation, formatting retrieved passages with passage IDs, and prompting Llama-3.2-1B-Instruct to decompose questions into atomic constraints such as answer type, target entity or event, relation, temporal or location qualifier, and negation or exclusion conditions.

**Methods.** I will first implement the required default project components. For Part 1, I will complete the GPT-small implementation, including multi-head self-attention with grouped query attention, RoPE, RMSNorm, the feed-forward network, transformer layers, attention masking, and causal language modeling loss. I will then run pretraining and finetune GPT-small on the required downstream tasks. For Part 2, I will implement the baseline RAG system, including BM25 retrieval, prompt construction from retrieved passages, and retrieval-augmented generation.

My proposed method is **constraint-violation-aware evidence filtering**. The method consists of four stages. First, BM25 retrieves the top- $k$  candidate passages for each question. Second, Llama-3.2-1B-Instruct decomposes the question into a small set of atomic constraints, such as answer type, target entity, relation, temporal qualifier, or negation condition. Third, for each retrieved passage, Llama predicts a constraint-wise diagnostic vector indicating whether each constraint is satisfied, missing, contradicted, or unrelated. Finally, passages are ranked using both their BM25 retrieval score and their diagnostic labels:

$$s(p) = \alpha B(p) + \beta S(p) - \gamma M(p) - \delta C(p),$$

where  $B(p)$  denotes the BM25 score,  $S(p)$  is the number of satisfied constraints,  $M(p)$  is the number of missing constraints, and  $C(p)$  is the number of contradicted constraints. Only passages with low violation scores and strong answer support are included in the final prompt for Llama-3.2-1B-Instruct. The final prompt instructs the model to generate an answer-only JSON object so that evaluation is not affected by unnecessary explanations or formatting variations.

The original part of this project is not the use of constraints in general, since QUEST already motivates constraint matching for retrieval. Rather, the original part is applying a lightweight, text-only constraint-violation diagnostic step to the default NQ RAG pipeline, especially for filtering hard-negative evidence before generation. I will implement the RAG subclass, prompt templates, diagnostic parsing code, scoring function, and evaluation scripts myself. The pretrained Llama-3.2-1B-Instruct model, tokenizer, Pyserini BM25 index (Lin et al., 2021), and NQ data will be downloaded or used as provided by the default project.

**Baselines.** I will compare the proposed method against several baselines. The first baseline is the naive BM25-based RAG system, where the top retrieved passages are directly inserted into the prompt with minimal prompt engineering. The second baseline is the prompt-tuned Llama-3.2-1B-Instruct RAG system, using the conversation-style input formatting and answer parsing required in the default project. This corresponds to the baseline RAG system in the default project handout. As a reference point, the default handout reports a zero-shot Llama-3.2-1B-Instruct RAG result of approximately 0.253 accuracy, 0.153 ROUGE-1, 0.071 ROUGE-2, and 0.151 ROUGE-L on the NQ development set, which I will compare against after reproducing the baseline. I will also compare with a relevance-only filtering baseline, where passages are selected only by BM25 score or by a simple Llama-predicted relevance score without constraint-wise labels. This will help determine whether the proposed diagnostic vector provides value beyond scalar relevance.

**Evaluation.** For quantitative evaluation, I will use the automatic metrics provided in the default project: exact-match-style accuracy and ROUGE scores, including

ROUGE-1, ROUGE-2, ROUGE-L, and ROUGE-Lsum. The main comparison will be between the default Llama RAG baseline and the proposed constraint-violation-aware filtering method on the NQ development set. I expect the proposed method to be most helpful on examples where BM25 retrieves hard negatives or partially relevant passages.

In addition to the standard NQ evaluation, I will construct a robustness evaluation using hard-negative injection. For each example, I will compare performance when the final prompt contains only top BM25 passages versus when it contains a mixture of positive or BM25-retrieved passages and hard negative contexts. I will measure both final QA accuracy and the performance drop as the number of hard negatives increases. A successful method should reduce the accuracy drop under hard-negative noise while using fewer or cleaner passages in the final prompt.

For qualitative analysis, I will inspect examples where the proposed method improves or hurts performance. I will categorize errors by the type of constraint involved, such as entity mismatch, relation mismatch, temporal mismatch, location mismatch, or negation failure. This analysis will help determine whether constraint-violation diagnostics are actually filtering misleading evidence, or whether the diagnostic prompts themselves introduce new errors.

## References

- Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. 2019. Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:452–466.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.
- Jimmy Lin, Xueguang Ma, Sheng-Chieh Lin, Jheng-Hong Yang, Ronak Pradeep, and Rodrigo Nogueira. 2021. Pyserini: A python toolkit for reproducible information retrieval research with sparse and dense representations. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2356–2362.
- Chaitanya Malaviya, Peter Shaw, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2023. [QUEST: A retrieval dataset of entity-seeking queries with implicit set operations](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14032–14047. Association for Computational Linguistics.
- Tri Nguyen, Mir Rosenberg, Xia Song, Jianfeng Gao, Saurabh Tiwary, Rangan Majumder, and Li Deng. 2016. MS MARCO: A human generated machine reading comprehension dataset. In *Proceedings of the Workshop on Cognitive Computation: Integrating Neural and Symbolic Approaches*.